

Project Requirement Document

Project : Rolling Robot

Last Updated : 29th August 2025

Requirements:

The Rolling-Robot must consist of all following functionalities during delivery :

- All features of the existing open-source project ESP-ROLL-Robot must function :

References :

<https://www.youtube.com/watch?v=VuBfRYNQgw0&t=1304s>

- In addition it needs to have the capability to do IMAGE-based-QA (with OpenAI).

The DF-ROBOT open-source code needs to be used and integrated with the ESP-ROLL main code :

References:

https://wiki.dfrobot.com/SKU_DFR1154_ESP32_S3_AI_CAM#6.8%20OpenAI%20Image%20Q&A

- Modify the ESP-ROLL PCB design to add the following modules:

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### Modifications
- **Microphone**: Use built-in PDM mic (GPIO 39 data, 38 clock). Optional:
  MAX4466 electret
  (
  [Amazon](https://www.amazon
  .com/MAX4466-Microphone-Amplifier-Adjustable-Arduino/dp/B07K8R4B1V)) with
  LM358 amp, non-conflicting ADC GPIO.
- **Speaker**: 8Ω, 0.5-1W, 15mm, MX1.25mm
  (
  [DigiKey](https://www.digikey
  .com/en/products/detail/visaton-gmbh-co-kg/FRS-5X-80HM/10406062)).
  MAX98357 amp (GPIO 42 DOUT, 45 BCLK, 46 LRC).
- **Laser Pointer**: 5mW, 650nm diode
  (
  [Amazon](https://www.amazon
  .com/KY-008-Laser-Transmitter-Module-Arduino/dp/B07M6L3V7C)), 100-200Ω
  resistor, SI2300 switch, GPIO 7 (PWM).
- **Distance Sensor**: VL53L0X ToF
  ([Adafruit](https://www.adafruit.com/product/3317)), I2C (GPIO 8 SDA, 9
  SCL), 4.7kΩ pull-ups.
- **Temperature Sensor**: MLX90614 IR
  ([SparkFun](https://www.sparkfun.com/products/9570)), I2C (GPIO 8 SDA, 9
  SCL), shared with VL53L0X.
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- Build a Test firmware to log data from the additional PCBA modules, essentially to test & ensure that the schematic is functioning fine.
- Modify the ESP-ROLL Mechanical design to add a BB8 like head.

Delivery :

- 10 working prototypes with all the above functions tested and verified.

Challenges :

The following are the list of challenges we foresee during the development process of the product :

FIRMWARE

- Integration of both the firmwares namely ESP-ROLL and DF-ROBOT OPEN-AI IMAGE-QA may be possible in one of the following ways depending on the feasibility :
 - o Switching between both the features using a physical hardware button on the robot.
 - o Switching between both the features using a toggle button on the web-app.
 - o Both the features working simultaneously without requiring any switching.

We may have to choose any of the above options based on the feasibility.

- Making the DF-ROBOT OPEN-AI IMAGE-QA firmware work based on its intended functionality. Since open-source codes may have bugs and issues.

HARDWARE

- Footprint of the modules need to fit within the available space.
- Availability of the modules in domestic sites.

MECHANICAL :

- Adding a BB8-like-Head mechanism can be challenging, wrt to the current timeline. It can be mitigated by using off-the-shelf parts/components.

Risks :

Certain risks are associated with the product's outcome and are listed below.

FIRMWARE

- The open source codes being used in this project are not tested by us, it may have bugs and issues - that may hinder our delivery timelines.

PRODUCT-DESIGN

- Mic and Speaker issues due to enclosure. Mic would be not able to pick up audio data properly. The output from the speaker might be affected and the user might not be able to hear audio from it clearly.